



VICTORIA JUNIOR COLLEGE
BIOLOGY DEPARTMENT
JC2 PRELIMINARY EXAMINATIONS 2014
Higher 2

VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE
 VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE
 VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE
 VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE

CANDIDATE NAME

CLASS

--	--	--	--	--

INDEX NUMBER

--	--

BIOLOGY

Paper 3

9648/03**18 September 2014**

Additional Materials: Answer Booklet/ Paper

2 Hours**READ THESE INSTRUCTIONS FIRST**

Write your CT GP/ INDEX NO. and name on all the work you hand in.

Write in dark blue or blue pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use any staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.

The intended number of marks is given in brackets [] at the end of each question.

For Examiner's Use	
Section A	
1	
2	
3	
Planning Question	
5	
Total	

Answer **all** questions

Question 1

- (a) Restriction fragment length polymorphism, RFLP, refers to the differences in restriction fragment length generated by restriction enzymes. Explain two ways in which the approach employed during RFLP analysis for the process of disease detection and for DNA fingerprinting differ.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (b) An autosomal recessive disease is caused by a mutation that causes a loss of a *BamHI* restriction within the gene locus as shown in Fig 1.1. The diagram shows the two different alleles of the gene being detected by a gene probe. (not drawn to scale)

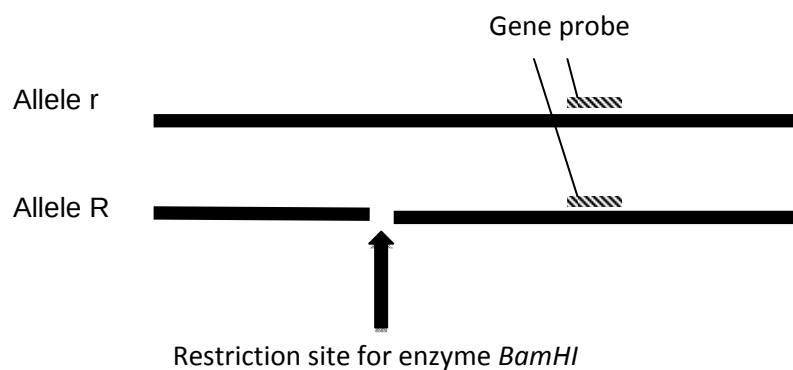


Fig. 1.1

- (i) Explain how this gene probe can be used to detect which individuals are carriers for the disease.

.....

.....

.....

.....

.....

..... [3]

- (c) A particular region where there is variable number of tandem repeats of a CGG trinucleotide has been proposed to be a good RFLP marker to detect another disease where the gene sequence is unknown. The unique sequence outside the repeat region is as follows:

5' CAGTATGCA------(CGG)_n-----ATGCGTAAT 3'
 3' GTCATACGT------(GCC)_n-----TACGCATTA 5'

- (i) Suggest two factors that are required in order for a RFLP marker to be useful in detection of the disease.

.....

.....

.....

..... [2]

- (ii) Explain why it is necessary to know the sequence outside the repeat region in order to use it during RFLP analysis for the process of disease detection.

.....

.....

.....

..... [2]

(d) Shown in Fig 1.3 is the outline of a pedigree for a certain X-linked dominant disease and a representation of a gel showing PCR-amplified fragments detecting the number of AAGGTT repeats in the RFLP marker used as a proxy for disease detection. Repeats of more than 42 at the RFLP marker region are associated with the disease. The DNA corresponding to each individual is directly below his or her place in the pedigree.

(i) Based on the information you have been given, fill in the pedigree to show the sex of the individuals I-1, I-2, II-3, II-4, as well as their phenotype (affected vs. unaffected). [2]

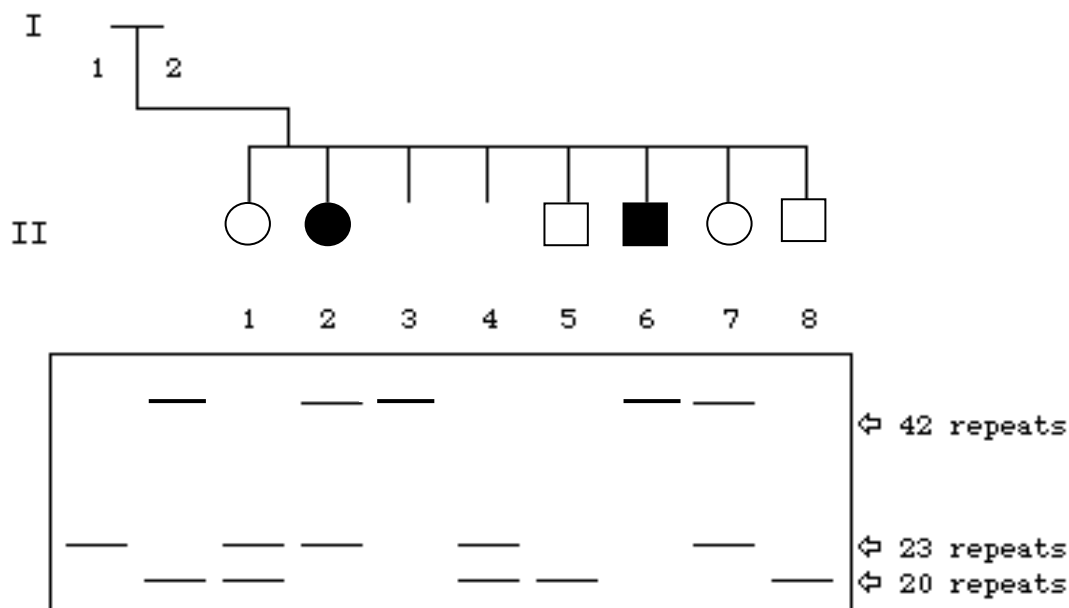


Fig. 1.3

(ii) In which individual did crossing over take place between the RFLP marker and the gene loci?

..... [1]

(iii) Explain your answer.

.....

 [3]

[Total: 17]

Question 2

Severe combined immunodeficiency (SCID) is a genetic disorder characterised by the absence of functional T-lymphocytes, resulting in the impairment of the immune system. There are different forms of SCID, each caused by mutation to a different gene. Two examples are X-linked SCID and ADA-SCID.

- (a) Compare the differences between the two forms of SCID mentioned in the paragraph above.

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (b) ADA-SCID can be treated using *Ex vivo* gene therapy. T-lymphocytes from the patient are isolated from the patient's blood. Retroviruses are then used to introduce the normal allele into the T-lymphocytes to allow normal development. These T-lymphocytes are then introduced back into the patient's blood. This process needs to be repeated every few months in order to prevent the re-occurrence of the disease.

- (i) Based on your knowledge of the structure of a retrovirus, explain why it is an serves as an efficient vector of gene therapy.

.....

.....

.....

..... [2]

- (ii) Suggest an advantage in carrying out *Ex vivo* gene therapy in the treatment of ADA-SCID.

.....
..... [1]

- (iii) Besides the need for repeated treatment, explain the other factors that keep this method of gene therapy from becoming an effective treatment for ADA-SCID.

.....
.....
.....
.....
..... [2]

- (c) In order to reduce the need for repeated treatment, a new method of gene therapy has been developed, in which the normal allele is introduced into blood stem cells instead of T-lymphocytes.

Distinguish between a blood stem cell and a T-lymphocyte.

.....
.....
.....
.....
.....
..... [3]

[Total: 12]

Question 3

Growth hormone genes from human, bovine or salmonid are introduced to farmed Atlantic salmon eggs to form transgenic fish that results in 10-80% increased growth rate than non-transgenic fish. Furthermore, fusing the gene with a strong gene promoter (e.g. ocean pout antifreeze promoter) further increases growth by 5 to 30 folds.

- (a) State one method of introducing the recombinant vector containing the transgene into the fish egg.

..... [1]

The company behind the genetically modified (GM) Atlantic salmon believes that it now stands on the verge of an historic decision by the powerful US Food and Drug Administration (FDA) that will open the way to the sale of genetically engineered meat and fish both in the United States and the rest of the world.

- b) (i) Explain how the approval by FDA may benefit the world.

.....

..... [1]

However, a recent study on farmed genetically modified Atlantic salmon (*Salmo salar*) observed a high incidence of spinal deformities in the fish. This could result in production losses. Many environmentalist and consumer groups are also protesting against the sale of the GM fish.

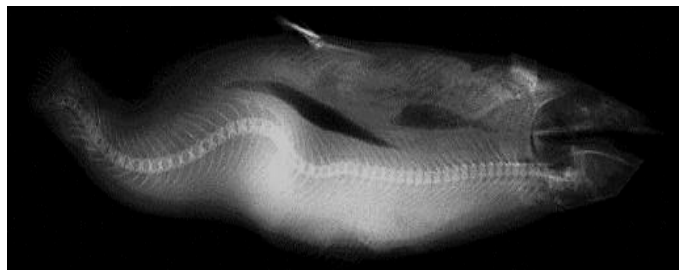


Fig 3.1: Structural deformities in farmed GM salmon

- (ii) With reference to Fig 3.1, suggest a likely ethical reason for the protest.

.....

..... [1]

- c) In response to the protest, the company issued a public statement that the deformities are not due to the genetic modifications but on the presence of cadmium ions in the local water supply. These ions affected the expression of a group of regulatory genes (VCR1, 2, 3, 4 and 5), known to be involved in the development of the fish.

The level of mRNA of the different genes in the farmed GM salmon with and without treatment of cadmium ions was measured in specific regions of the fish body.

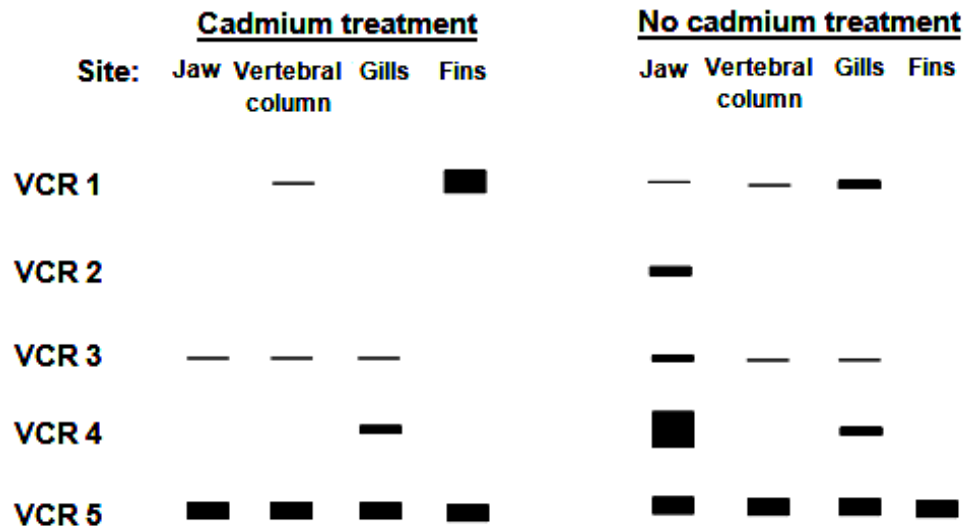


Fig 3.2

- (i) With reference to Fig 3.2, describe the effect of cadmium on the expression of the VCR genes for jaw development.

.....

.....

.....

..... [2]

- (ii) Base on Fig 3.2 and the information provided, explain if the spinal deformities in the GM fish is likely to be due to the effect of cadmium.

.....

.....

.....

..... [2]

- (iii)** A cDNA library for all the VCR genes was created for further studies. Explain which specific part of the fish was initially used as the starting biological sample.

.....

.....

.....

..... [2]

- d)** The company claimed that all its GM salmons are genetically engineered to be sterile. Explain why this will reduce the social implication of the GM organism.

.....

.....

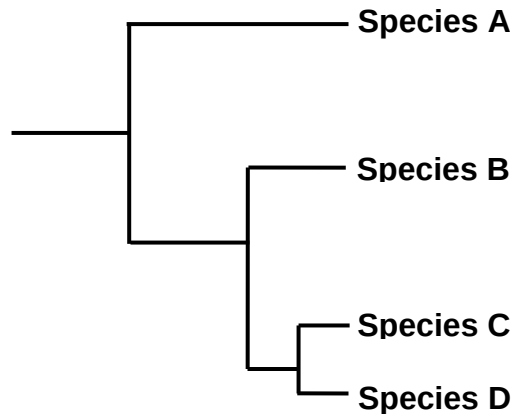
.....

..... [2]

Planning Question

4. The Orchidaceae, known as the orchid family, underwent a spectacularly diverse radiation since its late Cretaceous origin 83-75 million years ago to more than 27000 different species today. Classifying orchid species based on their evolutionary relationship is difficult and dynamic because of the large diversity of orchid species.

The phylogenetic tree of the four known Orchid species (A, B, C and D) is shown.



Using this information and your own knowledge, design an experiment to show the molecular homology between four species of orchids.

Your planning must be based on the assumption that you have been provided with the following equipment and materials.

- Plant tissue samples of the four orchid species
- Pestle and mortar
- DNA extraction buffer solution
- Glass rods
- Microfuge tubes
- Centrifuge machine
- Restriction enzyme, HindIII
- Suitable reagents for polymerase chain reaction
- DNA primers
- Thermal cycler
- Agarose gel plate
- Suitable source of electrical current

Your plan should have a clear and helpful structure to include:

- An explanation of the theory to support your practical procedure,
- a description of the method used,
- relevant, clearly labelled diagram,
- relevant risk and precaution taken.
- The correct use of technical and scientific terms

[Total 12]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

Free-response question

Write your answers to this question on the separate paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate.
- must be in continuous prose, where appropriate.
- must be set out in section (a), (b), etc., as indicated in the question.

- 5 (a) Outline the procedure for cloning the human insulin gene using E.coli cells to produce human insulin for diabetic patients. Your answer should include how the eukaryotic gene is cloned to avoid the problems associated with introns. [10]
- (b) Describe the advantages and limitations of PCR. [6]
- (c) Discuss the ethical and social considerations of gene therapy [4]

[Total: 20]